MATH 128 - HW11 - Jie Wen

Problem 1: Problems from the textbook, chapter 4.3

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(a) $\int_{-0.25}^{0.25} (\cos x)^2 dx$

Let $f(x) = \cos^2 x$, with limits $a = -0.25$ and $b = 0.25$. The step size is $h = 0.25 - (-0.25) = 0.5$.

The Trapezoidal approximation is:

$$T_a = \frac{h}{2}[f(-0.25) + f(0.25)] = \frac{0.5}{2}\left[\cos^2(-0.25) + \cos^2(0.25)\right]. = \frac{0.5}{2}[2\cos^2(0.25)] = 0.5\cos^2(0.25) \approx 0.4694.$$

(b) $\int_{-0.5}^{0} x\ln(x+1)dx$

Let $f(x) = x\ln(x+1)$, with limits $a = -0.5$ and $b = 0$. The step size is $h = 0 - (-0.5) = 0.5$.

The endpoints are evaluated as follows:

$$f(0) = 0 \cdot \ln 1 = 0,$$
$$f(-0.5) = -0.5\ln(0.5) = \frac{1}{2}\ln 2.$$

Applying the formula:

$$T_b = \frac{h}{2}[f(-0.5) + f(0)] = \frac{0.5}{2}\left[\frac{1}{2}\ln 2 + 0\right] = 0.125\ln 2 \approx 0.0866.$$

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(a) $\int_{-0.25}^{0.25} (\cos x)^2\, dx$

Use the formula $|E| \le \frac{h^3}{12}\max_{[a,b]}|f''(x)|$.

$$f'(x) = -\sin(2x), \quad f''(x) = -2\cos(2x).$$

Thus, $|f''(x)| = 2|\cos(2x)|$. On $[-0.25, 0.25]$, $2x \in [-0.5, 0.5]$ and $\cos(2x) \in [\cos(0.5), 1]$ (all positive), so the maximum is $2 \cdot 1 = 2$ at $x = 0$. Then:

$$|E| \le \frac{(0.5)^3}{12} \cdot 2 = \frac{0.125}{12} \cdot 2 \approx 0.020833333333333332.$$

The absolute actual error (0.02031712882950832) is less than the bound (0.020833333333333332), as expected.

(b) $\int_{-0.5}^{0} x\ln(x+1)\, dx$

Compute :

$$f'(x) = \ln(x+1) + \frac{x}{x+1}, \quad f''(x) = \frac{1}{x+1} + \frac{1}{(x+1)^2}.$$

On $[-0.5, 0]$, $f''(x) > 0$ and decreasing (as $x+1$ increases from 0.5 to 1), so the maximum is at $x = -0.5$: $f''(-0.5) = 2 + 4 = 6$. Then:

$$|E| \le \frac{(0.5)^3}{12} \cdot 6 = \frac{0.125}{12} \cdot 6 \approx 0.0625.$$

The absolute actual error (0.03407359027997266) is less than the bound (0.0625), as expected.

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6 & 8

(a) $\int_{-0.25}^{0.25} \cos^2 x\, dx$ **using Simpson's rule**

Simpson's rule (single application, n=2, two subintervals):

$$\int_a^b f(x)\, dx \approx \frac{h}{3}[f(a) + 4f(m) + f(b)], \quad h = \frac{b-a}{2}$$

Here $a = -0.25$, $b = 0.25$, so $h = 0.25$, midpoint $m = 0$.

$$f(-0.25) = \cos^2(-0.25) = \cos^2(0.25), \quad f(0) = \cos^2 0 = 1, \quad f(0.25) = \cos^2(0.25)$$

$$\frac{0.25}{3}\left[\cos^2(0.25) + 4 \cdot 1 + \cos^2(0.25)\right] = \frac{0.25}{3}\left[2\cos^2(0.25) + 4\right] = \frac{0.5}{3}\left[\cos^2(0.25) + 2\right]$$

Numerical value: $\cos^2(0.25) \approx 0.9387912809451864$

Approximation ≈ 0.4895964269817955

Exact value (as computed previously):

$$\int_{-0.25}^{0.25} \cos^2 x \, dx = 0.25 + \frac{1}{2}\sin(0.5) \approx 0.4897127693021015$$

Actual error ≈ 0.4895964269817955 − 0.4897127693021015 = **−0.0001163423203060**
Absolute actual error ≈ **1.163 × 10⁻⁴**

Simpson's error bound

$$|E| \le \frac{(b-a)^5}{2880} \max_{\xi \in [a,b]} |f^{(4)}(\xi)|$$

Fourth derivative of $f(x) = \cos^2 x$:
$f^{(4)}(x) = 8\cos(2x)$ (exact calculation)
$|f^{(4)}(x)| = 8|\cos(2x)| \le 8 \cdot 1 = 8$ on the interval.

$$|E| \le \frac{(0.5)^5}{2880} \cdot 8 = \frac{0.03125}{2880} \cdot 8 = \frac{0.25}{2880} \approx 8.6806 \times 10^{-5}$$

Bound ≈ **8.68 × 10⁻⁵**, actual error ≈ **1.16 × 10⁻⁴**

Part b: $\int_{-0.5}^{0} x\ln(x+1)\,dx$ using Simpson's rule

Same points: $a = -0.5$, $b = 0$, $h = 0.25$, midpoint $m = -0.25$

$$f(-0.5) = -0.5\ln(0.5) = 0.5\ln 2 \approx 0.34657359027997264$$

$$f(-0.25) = -0.25\ln(0.75) \approx -0.25 \times 0.2876820724517809 \approx -0.07192051811294522$$

$$f(0) = 0$$

Simpson approximation:

$$\frac{0.25}{3}[0.34657359027997264 + 4(-0.07192051811294522) + 0] \approx \frac{0.25}{3}(0.34657359027997264 - 0.2876820724517809) \approx 0.05256383927606418$$

Exact value (from integration by parts, previously computed) ≈ **0.05256980729002051**

Actual error ≈ 0.05256383927606418 − 0.05256980729002051 = **−0.00000596801395633**
Absolute actual error ≈ **5.97 × 10⁻⁶**

Simpson's error bound
Need $|f^{(4)}(x)|$ for $f(x) = x\ln(x+1)$.

Derivatives (computed exactly):

$$f''(x) = \frac{1}{x+1} + \frac{1}{(x+1)^2} = \frac{2x+3}{(x+1)^3}$$

$$f'''(x) = -\frac{2}{(x+1)^3} - \frac{6}{(x+1)^4} = -\frac{2(x+2)}{(x+1)^4}$$

$$f^{(4)}(x) = \frac{8}{(x+1)^4} + \frac{24}{(x+1)^5} = \frac{8(x+1)+24}{(x+1)^5} = \frac{8x+32}{(x+1)^5}$$

On $[-0.5, 0]$, $x+1 \in [0.5, 1]$, so denominator is positive and decreasing; numerator is positive.
Maximum of $|f^{(4)}(x)|$ occurs at left endpoint $x = -0.5$:

$$f^{(4)}(-0.5) = \frac{8(-0.5)+32}{(0.5)^5} = \frac{28}{0.03125} = 896$$

Thus $\max|f^{(4)}| = 896$.

Error bound:

$$|E| \le \frac{(0.5)^5}{2880} \cdot 896 = \frac{0.03125 \times 896}{2880} = \frac{28}{2880} = \frac{7}{720} \approx 0.00972222$$

Bound ≈ **0.00972**, actual error ≈ **5.97 × 10⁻⁶**

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The relationship between these three rules is applied on the same interval with matching step sizes.

On $[0, 2]$ with step $h = 1$, the composite rules are:

- **Trapezoidal rule** (with nodes $0, 1, 2$):

$$T = \frac{h}{2}[f(0) + 2f(1) + f(2)] = \frac{1}{2}[f(0) + 2f(1) + f(2)].$$

- **Midpoint rule** over $[0, 2]$ (single midpoint $x = 1$):

$$M = 2f(1).$$

- **Simpson's rule** (same three nodes $0, 1, 2$):

$$S = \frac{h}{3}[f(0) + 4f(1) + f(2)] = \frac{1}{3}[f(0) + 4f(1) + f(2)].$$

S can be expressed as a linear combination of T and M:

$$S = aT + bM$$

for some constants a, b. Matching coefficients of $f(0), f(1), f(2)$ gives

$$a = \frac{2}{3}, \quad b = \frac{1}{3},$$

so

$$S = \frac{2}{3}T + \frac{1}{3}M.$$

Plugging in the given values $T = 5$ and $M = 4$:

$$S = \frac{2}{3} \cdot 5 + \frac{1}{3} \cdot 4 = \frac{10 + 4}{3} = \frac{14}{3} \approx 4.6667.$$

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This is essentially the 2-point Gauss–Legendre rule on $[-1, 1]$, whose nodes are the roots of $P_2(x) = \frac{1}{2}(3x^2 - 1)$, i.e. $\pm\frac{\sqrt{3}}{3}$. For n Gauss–Legendre nodes, the degree of precision is $2n - 1$; for $n = 2$ that's 3.

Testing monomials directly:

The formula is tested for exactness using $f(x) = x^k$:

$$\int_{-1}^{1} x^k\,dx \stackrel{?}{=} \left(-\frac{\sqrt{3}}{3}\right)^k + \left(\frac{\sqrt{3}}{3}\right)^k.$$

Check $k = 0, 1, 2, 3$

1. $k = 0, f(x) = 1$**:**

$$\int_{-1}^{1} 1\,dx = 2, \quad 1 + 1 = 2 \quad \checkmark$$

2. $k = 1, f(x) = x$**:**

$$\int_{-1}^{1} x\,dx = 0, \quad -\frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{3} = 0 \quad \checkmark$$

3. $k = 2, f(x) = x^2$**:**

$$\int_{-1}^{1} x^2\,dx = \frac{2}{3}.$$

   Right-hand side:

$$\left(-\frac{\sqrt{3}}{3}\right)^2 + \left(\frac{\sqrt{3}}{3}\right)^2 = 2 \cdot \frac{3}{9} = \frac{2}{3} \quad \checkmark$$

4. $k = 3, f(x) = x^3$**:**

$$\int_{-1}^{1} x^3\,dx = 0,$$

   and

$$\left(-\frac{\sqrt{3}}{3}\right)^3 + \left(\frac{\sqrt{3}}{3}\right)^3 = -\frac{\sqrt{3}^3}{27} + \frac{\sqrt{3}^3}{27} = 0 \quad \checkmark$$

The rule is exact for $k = 0, 1, 2, 3$.

Check $k = 4$

$$\int_{-1}^{1} x^4, dx = \frac{2}{5}.$$

Right-hand side:

$$\left(-\frac{\sqrt{3}}{3}\right)^4 + \left(\frac{\sqrt{3}}{3}\right)^4 = 2 \cdot \left(\frac{3}{9}\right)^2 = 2 \cdot \frac{1}{9} = \frac{2}{9}.$$

$\frac{2}{5} \neq \frac{2}{9}$, so the rule is **not** exact for degree 4.

Therefore, the largest integer n such that the formula is exact for all polynomials of degree $\leq n$ is $n = 3$, so the **degree of precision is 3**.

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The formula is tested on monomials ($f(x) = 1, x, x^2, \ldots$) until it stops being exact.

- $h = \dfrac{b-a}{3}$
- $x_0 = a$, $x_1 = a + h$, $x_2 = b = a + 3h$
- Quadrature:

$$Q(f) = \frac{9}{4}hf(x_1) + \frac{3}{4}hf(x_2)$$

1. Test $f(x) = 1$

Exact integral:

$$\int_a^b 1, dx = b - a = 3h$$

Quadrature:

$$Q(1) = \frac{9}{4}h \cdot 1 + \frac{3}{4}h \cdot 1 = \left(\frac{9}{4} + \frac{3}{4}\right)h = \frac{12}{4}h = 3h$$

$\Rightarrow$ Exact for x^0.

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2. Test $f(x) = x$

Exact integral:

$$\int_a^b x, dx = \frac{b^2 - a^2}{2} = \frac{(a+3h)^2 - a^2}{2} = \frac{a^2 + 6ah + 9h^2 - a^2}{2} = 3ah + \frac{9}{2}h^2$$

Quadrature:

$$Q(x) = \frac{9}{4}h(a + h) + \frac{3}{4}h(a + 3h)$$

$$= \frac{9}{4}ha + \frac{9}{4}h^2 + \frac{3}{4}ha + \frac{9}{4}h^2 = \left(\frac{9}{4} + \frac{3}{4}\right)ha + \left(\frac{9}{4} + \frac{9}{4}\right)h^2 = 3ah + \frac{9}{2}h^2$$

$\Rightarrow$ Exact for x^1.

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3. Test $f(x) = x^2$

Exact integral:

$$\int_a^b x^2, dx = \frac{b^3 - a^3}{3} = \frac{(a+3h)^3 - a^3}{3}$$

$$(a+3h)^3 = a^3 + 9a^2h + 27ah^2 + 27h^3$$

So

$$\int_a^b x^2, dx = \frac{a^3 + 9a^2h + 27ah^2 + 27h^3 - a^3}{3} = 3a^2h + 9ah^2 + 9h^3$$

Quadrature:

$$Q(x^2) = \frac{9}{4}h(a+h)^2 + \frac{3}{4}h(a+3h)^2$$

$$(a+h)^2 = a^2 + 2ah + h^2, \quad (a+3h)^2 = a^2 + 6ah + 9h^2$$

So

$$Q(x^2) = \frac{9}{4}h(a^2 + 2ah + h^2) + \frac{3}{4}h(a^2 + 6ah + 9h^2)$$

$$= \left(\frac{9}{4} + \frac{3}{4}\right)ha^2 + \left(\frac{9}{2} + \frac{18}{4}\right)h^2a + \left(\frac{9}{4} + \frac{27}{4}\right)h^3$$

$$= 3a^2h + 9ah^2 + 9h^3$$

$\Rightarrow$ Exact for x^2.

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4. Test $f(x) = x^3$

Exact integral:

$$\int_a^b x^3, dx = \frac{b^4 - a^4}{4} = \frac{(a+3h)^4 - a^4}{4}$$

$$(a+3h)^4 = a^4 + 12a^3h + 54a^2h^2 + 108ah^3 + 81h^4$$

So

$$\int_a^b x^3, dx = \frac{12a^3h + 54a^2h^2 + 108ah^3 + 81h^4}{4}$$

Quadrature:

$$Q(x^3) = \frac{9}{4}h(a+h)^3 + \frac{3}{4}h(a+3h)^3$$

$$(a+h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$$

$$(a+3h)^3 = a^3 + 9a^2h + 27ah^2 + 27h^3$$

So

$$Q(x^3) = \frac{9}{4}h(a^3 + 3a^2h + 3ah^2 + h^3) + \frac{3}{4}h(a^3 + 9a^2h + 27ah^2 + 27h^3)$$

Computing coefficients:

- a^3h: $\frac{9}{4} + \frac{3}{4} = 3$
- a^2h^2: $\frac{27}{4} + \frac{27}{4} = \frac{54}{4}$
- ah^3: $\frac{27}{4} + \frac{81}{4} = \frac{108}{4}$
- h^4: $\frac{9}{4} + \frac{81}{4} = \frac{90}{4}$

So

$$Q(x^3) = 3a^3h + \frac{54}{4}a^2h^2 + \frac{108}{4}ah^3 + \frac{90}{4}h^4$$

Comparison with exact:

$$\int_a^b x^3, dx = 3a^3h + \frac{54}{4}a^2h^2 + \frac{108}{4}ah^3 + \frac{81}{4}h^4$$

They differ in the h^4 term ($\frac{90}{4}$ vs $\frac{81}{4}$), so the rule is **not** exact for x^3.

$\Rightarrow$ Not exact for x^3.

Conclusion

The formula is exact for x^0, x^1, x^2, but not for x^3.

So the **degree of precision** of the quadrature formula is: $n = 2$

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Since the formula is exact for all polynomials of degree ≤ 2, it must be exact for the basis polynomials $f(x) = 1, x, x^2$ on $[0, 2]$.

The quadrature formula is:

$$\int_0^2 f(x)dx = c_0 f(0) + c_1 f(1) + c_2 f(2).$$

Use $f(x) = 1$:

$$c_0 + c_1 + c_2 = 2. \tag{1}$$

Use $f(x) = x$:

$$c_1 + 2c_2 = 2. \tag{2}$$

Use $f(x) = x^2$:

$$c_1 + 4c_2 = \frac{8}{3}. \tag{3}$$

Solve the system:

$$c_0 = \frac{1}{3}, \quad c_1 = \frac{4}{3}, \quad c_2 = \frac{1}{3}$$

(So this is exactly Simpson's rule on $[0, 2]$ with step size $h = 1$: $\int_0^2 f(x)dx \approx \frac{1}{3}\big[f(0) + 4f(1) + f(2)\big]$.)

Problem 2: Problems from the textbook, chapter 4.4

1

For the Composite Trapezoidal rule with n equal subintervals on $[a, b]$,

$$h = \frac{b-a}{n}, \quad x_j = a + jh,$$
$$\int_a^b f(x), dx \approx T_n = h\left[\frac{1}{2}f(x_0) + \sum_{j=1}^{n-1} f(x_j) + \frac{1}{2}f(x_n)\right].$$

(a) $\int_1^2 x \ln x, dx, \quad n = 4$

- $a = 1,\ b = 2,\ n = 4 \Rightarrow h = \dfrac{2-1}{4} = 0.25.$
- Nodes: $x_j = 1 + 0.25j$:
  $x_0 = 1,\ x_1 = 1.25,\ x_2 = 1.5,\ x_3 = 1.75,\ x_4 = 2.$

$$\begin{aligned} T_4 &= h\Big[\tfrac{1}{2}f(1) + f(1.25) + f(1.5) + f(1.75) + \tfrac{1}{2}f(2)\Big] \\ &= 0.25\Big[0 + 0.27893 + 0.60820 + 0.97933 + 0.5(1.38629)\Big] \\ &\approx 0.63990. \end{aligned}$$

(d) $\int_0^\pi x^2 \cos x, dx, \quad n = 6$

- $a = 0,\ b = \pi,\ n = 6 \Rightarrow h = \dfrac{\pi - 0}{6} = \dfrac{\pi}{6}.$
- Nodes: $x_j = j\frac{\pi}{6},\ j = 0, \ldots, 6.$

$$\begin{aligned} T_6 &= h\Big[\tfrac{1}{2}f(0) + f(x_1) + \cdots + f(x_5) + \tfrac{1}{2}f(\pi)\Big] \\ &= \frac{\pi}{6}\Big[0 + 0.23743 + 0.54831 + 0 + (-2.19325) + (-5.93564) + \tfrac{1}{2}(-9.86960)\Big] \\ &\approx -6.42872. \end{aligned}$$

(e) $\int_0^2 e^{2x} \sin 3x, dx, \quad n = 8$

- $a = 0,\ b = 2,\ n = 8 \Rightarrow h = \dfrac{2-0}{8} = 0.25.$
- Nodes: $x_j = 0.25j,\ j = 0, \ldots, 8.$

$$\begin{aligned} T_8 &= h\Big[\tfrac{1}{2}f(0) + f(0.25) + f(0.5) + \cdots + f(1.75) + \tfrac{1}{2}f(2)\Big] \\ &= 0.25\Big[0 + 1.12383 + 2.71147 + 3.48708 + 1.04274 \\ &\qquad -6.96304 - 19.63422 - 28.44400 + \tfrac{1}{2}(-15.25557)\Big] \\ &\approx -13.57598. \end{aligned}$$

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The **Composite Simpson's rule** (with even n) is used:

$$\int_a^b f(x),dx \approx S_n = \frac{h}{3}\left[f(x_0)+f(x_n)+4\sum_{\substack{j=1\\ j\,\text{odd}}}^{n-1} f(x_j)+2\sum_{\substack{j=2\\ j\,\text{even}}}^{n-2} f(x_j)\right],$$

where $h=\dfrac{b-a}{n}$, $x_j=a+jh$.

(a) $\displaystyle\int_1^2 x\ln x,dx, \quad n=4$

- $a=1,\ b=2,\ n=4,\ h=\dfrac{2-1}{4}=0.25.$
- Nodes: $x_j=1+0.25j$, so
  $x_0=1,\ x_1=1.25,\ x_2=1.5,\ x_3=1.75,\ x_4=2.$

$$\begin{aligned} S_4 &= \frac{h}{3}\Big[f(x_0)+f(x_4)+4(f(x_1)+f(x_3))+2f(x_2)\Big] \\ &= \frac{0.25}{3}\Big[0+1.38629+4(0.27893+0.97933)+2(0.60820)\Big] \\ &\approx 0.63631. \end{aligned}$$

(d) $\displaystyle\int_0^{\pi} x^2\cos x,dx, \quad n=6$

- $a=0,\ b=\pi,\ n=6,\ h=\dfrac{\pi}{6}.$
- Nodes: $x_j=j\frac{\pi}{6},\ j=0,\dots,6.$

Composite Simpson:

$$\begin{aligned} S_6 &= \frac{h}{3}\Big[f(x_0)+f(x_6)+4(f(x_1)+f(x_3)+f(x_5))+2(f(x_2)+f(x_4))\Big] \\ &= \frac{\pi/6}{3}\Big[0-9.86960+4(0.23743+0-5.93564)+2(0.54831-2.19325)\Big] \\ &\approx -6.27487. \end{aligned}$$

(e) $\displaystyle\int_0^2 e^{2x}\sin 3x,dx, \quad n=8$

- $a=0,\ b=2,\ n=8,\ h=\dfrac{2-0}{8}=0.25.$
- Nodes: $x_j=0.25j,\ j=0,\dots,8.$

Composite Simpson:

$$\begin{aligned} S_8 &= \frac{h}{3}\Big[f(x_0)+f(x_8)+4(f(x_1)+f(x_3)+f(x_5)+f(x_7)) \\ &\qquad +2(f(x_2)+f(x_4)+f(x_6))\Big] \\ &= \frac{0.25}{3}\Big[0-15.25557+4(1.12383+3.48708-6.96304-28.44400) \\ &\qquad +2(2.71147+1.04274-19.63422)\Big] \\ &\approx -14.18334. \end{aligned}$$

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The integral

$$I=\int_1^2 x\ln x,dx$$

is computed with accuracy 10^{-5}.

$$f'(x)=\ln x+1, \quad f''(x)=\frac{1}{x}, \quad f^{(3)}(x)=-\frac{1}{x^2}, \quad f^{(4)}(x)=\frac{2}{x^3}.$$

On $[1,2]$:

- $\max|f''(x)|=f''(1)=1.$
- $\max|f^{(4)}(x)|=f^{(4)}(1)=2.$

Also, the exact value (for checking) is

$$I=\Big[\frac{x^2}{2}\ln x-\frac{x^2}{4}\Big]_1^2=2\ln 2-\frac{3}{4}\approx 0.63629436.$$

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a. Composite Trapezoidal Rule

The error formula (Theorem 4.5) is:

$$\int_a^b f(x),dx = T_n - \frac{b-a}{12}h^2 f''(\mu),$$

so

$$|E_T| \le \frac{b-a}{12}h^2 \max|f''(x)|.$$

Here $a=1, b=2 \Rightarrow b-a=1$, $\max|f''|=1$:

$$|E_T| \le \frac{1}{12}h^2.$$

Requiring $|E_T| \le 10^{-5}$:

$$\frac{1}{12}h^2 \le 10^{-5} \quad \Rightarrow \quad h \le 0.010954\ldots$$

With $h = \frac{b-a}{n} = \frac{1}{n}$, this means

$$\frac{1}{n} \le 0.010954 \ \Rightarrow \ n \ge 91.29.$$

So the smallest integer that guarantees the tolerance is

$$\boxed{n = 92, \quad h = \frac{1}{92}.}$$

The composite trapezoidal approximation is:

$$T_{92} = \frac{h}{2}\Big(f(1) + 2\sum_{j=1}^{91} f(1+jh) + f(2)\Big) \approx \boxed{0.63630119}.$$

Error vs exact:

$$T_{92} - I \approx 6.8 \times 10^{-6},$$

which is within 10^{-5}, as guaranteed.

b. Composite Simpson's Rule

The error formula (Theorem 4.4) is:

$$\int_a^b f(x),dx = S_n - \frac{b-a}{180}h^4 f^{(4)}(\mu),$$

so

$$|E_S| \le \frac{b-a}{180}h^4 \max|f^{(4)}(x)|.$$

Here $b-a=1$, $\max|f^{(4)}|=2$:

$$|E_S| \le \frac{1}{180}h^4 \cdot 2 = \frac{1}{90}h^4.$$

Requiring $|E_S| \le 10^{-5}$:

$$\frac{1}{90}h^4 \le 10^{-5} \quad \Rightarrow \quad h^4 \le 9 \times 10^{-4} \quad \Rightarrow \quad h \le 0.1732\ldots$$

Again $h = 1/n$, so

$$\frac{1}{n} \le 0.1732 \ \Rightarrow \ n \ge 5.77.$$

Simpson's rule requires **even** n, so take

$$\boxed{n = 6, \quad h = \frac{1}{6}.}$$

The composite Simpson approximation is:

$$S_6 = \frac{h}{3}\Big(f(1) + 2\sum_{j=1}^{2} f(1+2jh) + 4\sum_{j=1}^{3} f(1+(2j-1)h) + f(2)\Big) \approx \boxed{0.63629750}.$$

Error:

$$S_6 - I \approx 3.1 \times 10^{-6},$$

comfortably below 10^{-5}.

c. Composite Midpoint Rule

Using the form

$$\int_a^b f(x),dx = 2h\sum_{j=1}^{n/2} f(x_{2j-1}) + \frac{b-a}{6}h^2 f''(\mu),$$

with $a = x_0, b = x_n, h = (b-a)/n$, and n **even**.

Error bound:

$$|E_M| \le \frac{b-a}{6}h^2 \max|f''(x)|.$$

Here $b-a = 1$, $\max|f''| = 1$:

$$|E_M| \le \frac{1}{6}h^2.$$

Requiring $|E_M| \le 10^{-5}$:

$$\frac{1}{6}h^2 \le 10^{-5} \quad \Rightarrow \quad h^2 \le 6\times 10^{-5} \quad \Rightarrow \quad h \le 0.007746\ldots$$

With $h = 1/n$:

$$\frac{1}{n} \le 0.007746 \;\Rightarrow\; n \ge 129.10.$$

n must be even, so take

$$\boxed{n = 130, \quad h = \frac{1}{130}.}$$

The composite midpoint approximation (with midpoints $x_{2j-1} = 1 + (2j-1)h$, $j = 1,\ldots,65$) is:

$$M_{130} = 2h\sum_{j=1}^{65} f\big(1+(2j-1)h\big) \approx \boxed{0.63628753}.$$

Error:

$$M_{130} - I \approx -6.8\times 10^{-6},$$

again within the required 10^{-5}.

Problem 3: Nonlinear equations

Step 1: Case $|\phi'(x^*)| < 1$

If

$$|\phi'(x^*)| < 1,$$

then take $g = \phi$ in the standard fixed-point theorem.

- ϕ is continuously differentiable by assumption.
- x^* is a fixed point: $\phi(x^*) = x^*$.
- $|\phi'(x^*)| < 1$.

By continuity of ϕ', there exists a small neighborhood of x^* in which

$$|\phi'(x)| \le q < 1.$$

Thus, on that neighborhood, ϕ is a contraction, and **iteration (I)**

$$x_{k+1} = \phi(x_k)$$

is locally convergent to x^*.

So in this case, (I) converges.

Step 2: Case $|\phi'(x^*)| > 1$

Now suppose

$$|\phi'(x^*)| > 1.$$

Then look at iteration (II), whose iteration function is $g = \phi^{-1}$.

1. x^* is a fixed point of ϕ: $\phi(x^*) = x^*$.
   Applying ϕ^{-1} to both sides gives

   $$\phi^{-1}(x^*) = x^*,$$

   so x^* is also a fixed point of ϕ^{-1}. Thus for iteration (II),

   $$g(x) = \phi^{-1}(x), \quad g(x^*) = x^*.$$

2. We want $|g'(x^*)| < 1$.
   When ϕ is continuously differentiable and $\phi'(x^*) \neq 0$, we can use the derivative-of-inverse formula:

   $$(\phi^{-1})'(y) = \frac{1}{\phi'(x)} \quad \text{where } y = \phi(x).$$

   At the fixed point, $y = x^* = \phi(x^*)$, and $\phi^{-1}(x^*) = x^*$, so

   $$(\phi^{-1})'(x^*) = \frac{1}{\phi'(x^*)}.$$

   Hence

   $$\left|(\phi^{-1})'(x^*)\right| = \left|\frac{1}{\phi'(x^*)}\right| = \frac{1}{|\phi'(x^*)|}.$$

   Under the assumption $|\phi'(x^*)| > 1$, we get

   $$\left|(\phi^{-1})'(x^*)\right| = \frac{1}{|\phi'(x^*)|} < 1.$$

3. Since ϕ is C^1, its inverse is (locally) C^1 near x^* where defined, and its derivative is continuous. Thus, by continuity of $(\phi^{-1})'$, there is a neighborhood of x^* where

   $$\left|(\phi^{-1})'(x)\right| \leq q < 1.$$

   So ϕ^{-1} is a contraction on a neighborhood of x^*.

Therefore, by the fixed-point iteration theorem, **iteration (II)**

$$x_{k+1} = \phi^{-1}(x_k)$$

is locally convergent to x^*.

Step 3: The case $\phi'(x^*) = 0$

This is already covered by Step 1:

- If $\phi'(x^*) = 0$, then $|\phi'(x^*)| = 0 < 1$, so iteration (I) converges locally by the same contraction argument.
- The inverse is not needed in this case.

Step 4: Conclusion

Given that $|\phi'(x)| \neq 1$; in particular, at the fixed point x^*,

- either $|\phi'(x^*)| < 1$,
- or $|\phi'(x^*)| > 1$.

In the first case, iteration (I) converges locally. In the second case, $|\phi'(x^*)| > 1$ implies

$$\left|(\phi^{-1})'(x^*)\right| = \frac{1}{|\phi'(x^*)|} < 1,$$

so iteration (II) converges locally.

Thus **in all cases, at least one of the two iterations is locally convergent to the fixed point** x^*.

Problem 4: Interpolation

Let the three distinct interpolation points be a, b, c and write their function values as $f(a), f(b), f(c)$.

(a) Newton and Lagrange forms of the quadratic interpolant

Lagrange form

The quadratic interpolant $p(x)$ through $(a, f(a)), (b, f(b)), (c, f(c))$ is

$$p(x) = f(a)L_0(x) + f(b)L_1(x) + f(c)L_2(x),$$

where the Lagrange basis polynomials are

$$L_0(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)}, \quad L_1(x) = \frac{(x-a)(x-c)}{(b-a)(b-c)}, \quad L_2(x) = \frac{(x-a)(x-b)}{(c-a)(c-b)}.$$

Newton form

Using the nodes in the order $x_0 = a, x_1 = b, x_2 = c$, the Newton (divided-difference) form is

$$p(x) = f[a] + f[a,b](x-a) + f[a,b,c](x-a)(x-b),$$

where the divided differences are

$$f[a] = f(a), \quad f[a,b] = \frac{f(b)-f(a)}{b-a}, \quad f[b,c] = \frac{f(c)-f(b)}{c-b},$$

$$f[a,b,c] = \frac{f[b,c]-f[a,b]}{c-a}.$$

(b) Error formula and interpretation

General interpolation error for degree- n polynomial at nodes $x_0, \ldots, x_n$ is

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{j=0}^{n} (x - x_j)$$

for some ξ_x between the smallest and largest of $x, x_0, \ldots, x_n$.

Here $n = 2$ and $x_0 = a, x_1 = b, x_2 = c$, so

$$\boxed{f(x) - p(x) = \frac{f^{(3)}(\xi_x)}{3!}(x-a)(x-b)(x-c), \quad \xi_x \in (\min\{a,b,c,x\}, \max\{a,b,c,x\})}$$

Equivalently,

$$p(x) - f(x) = -\frac{f^{(3)}(\xi_x)}{6}(x-a)(x-b)(x-c).$$

Justification:

- **Zeroes:** At each interpolation node $x = a, b, c$, one factor in $(x-a)(x-b)(x-c)$ is zero, so the error is zero exactly where interpolation is forced.
- **Derivatives vs degree:** For a quadratic interpolant (degree 2) the next derivative $f^{(3)}$ controls the error. If f itself is a quadratic, then $f^{(3)} \equiv 0$ and the error is identically zero, as expected.
- **Dimensions (units):** Each $(x - \text{node})$ has units of x, so the product has units x^3. The third derivative $f^{(3)}$ has units "function / x^3". Their product has the same units as f, so the error has the right dimension.

(c) Newton coefficients for $f(x) = R - \dfrac{1}{x}$

The divided differences for

$$f(x) = R - \frac{1}{x}$$

at nodes a, b, c are computed as follows.

1. Zeroth order:

$$f[a] = R - \frac{1}{a}, \quad f[b] = R - \frac{1}{b}, \quad f[c] = R - \frac{1}{c}.$$

2. First order:

$$f[a,b] = \frac{f(b)-f(a)}{b-a} = \frac{\left(R - \frac{1}{b}\right) - \left(R - \frac{1}{a}\right)}{b-a} = \frac{-\frac{1}{b} + \frac{1}{a}}{b-a} = \frac{\frac{a-b}{ab}}{b-a} = -\frac{1}{ab}.$$

   Similarly,

$$f[b,c] = \frac{f(c)-f(b)}{c-b} = \frac{-\frac{1}{c} + \frac{1}{b}}{c-b} = -\frac{1}{bc}.$$

3. Second order:

$$f[a,b,c] = \frac{f[b,c]-f[a,b]}{c-a} = \frac{-\frac{1}{bc} - \left(-\frac{1}{ab}\right)}{c-a} = \frac{-\frac{1}{bc} + \frac{1}{ab}}{c-a} = \frac{\frac{1}{ab} - \frac{1}{bc}}{c-a} = \frac{\frac{c-a}{abc}}{c-a} = \frac{1}{abc}.$$

So the **Newton form** of the interpolant is

$$\boxed{p(x) = \left(R - \frac{1}{a}\right) - \frac{1}{ab}(x-a) + \frac{1}{abc}(x-a)(x-b).}$$

(d) Convert $p(x)$ to standard form $q_0 + q_1 x + q_2 x^2$

Given

$$p(x) = \left(R - \frac{1}{a}\right) - \frac{1}{ab}(x-a) + \frac{1}{abc}(x-a)(x-b).$$

Expanding step by step:

1. Expand the linear term:

$$-\frac{1}{ab}(x-a) = -\frac{1}{ab}x + \frac{1}{b}.$$

2. Expand the quadratic factor:

$$(x-a)(x-b) = x^2 - (a+b)x + ab.$$

So

$$\frac{1}{abc}(x-a)(x-b) = \frac{1}{abc}x^2 - \frac{a+b}{abc}x + \frac{ab}{abc} = \frac{1}{abc}x^2 - \frac{a+b}{abc}x + \frac{1}{c}.$$

3. Add constant, linear, quadratic pieces.

Constant term:

$$q_0 = \left(R - \frac{1}{a}\right) + \frac{1}{b} + \frac{1}{c} = R - \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

Coefficient of x:

$$q_1 = -\frac{1}{ab} - \frac{a+b}{abc} = -\frac{c}{abc} - \frac{a+b}{abc} = -\frac{a+b+c}{abc}.$$

Coefficient of x^2:

$$q_2 = \frac{1}{abc}.$$

Thus

$$\boxed{p(x) = q_0 + q_1 x + q_2 x^2 = \left(R - \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) - \frac{a+b+c}{abc}x + \frac{1}{abc}x^2.}$$

Problem 5: Numerical differentiation

$f'(x_1)$ is approximated by

$$D_0 f(x_1) := \frac{f(x_2) - f(x_0)}{2h}, \quad x_2 - x_1 = x_1 - x_0 = h > 0,$$

so $x_0 = x_1 - h$, $x_2 = x_1 + h$. This is the standard *central difference* formula.

Assume $f \in C^3$ on $[x_0, x_2]$.

Taylor's theorem with remainder is applied around x_1:

$$\begin{aligned} f(x_1 + h) &= f(x_1) + hf'(x_1) + \frac{h^2}{2}f''(x_1) + \frac{h^3}{6}f^{(3)}(\xi_2), \\ f(x_1 - h) &= f(x_1) - hf'(x_1) + \frac{h^2}{2}f''(x_1) - \frac{h^3}{6}f^{(3)}(\xi_0), \end{aligned}$$

for some $\xi_2 \in (x_1, x_1 + h)$, $\xi_0 \in (x_1 - h, x_1)$.

Subtracting the second equation from the first yields:

$$\begin{aligned} f(x_1 + h) - f(x_1 - h) &= \left[f(x_1) + hf'(x_1) + \tfrac{h^2}{2}f''(x_1) + \tfrac{h^3}{6}f^{(3)}(\xi_2)\right] \\ &\quad - \left[f(x_1) - hf'(x_1) + \tfrac{h^2}{2}f''(x_1) - \tfrac{h^3}{6}f^{(3)}(\xi_0)\right] \\ &= 2hf'(x_1) + \frac{h^3}{6}\left(f^{(3)}(\xi_2) + f^{(3)}(\xi_0)\right). \end{aligned}$$

Dividing by $2h$:

$$D_0 f(x_1) = \frac{f(x_1 + h) - f(x_1 - h)}{2h} = f'(x_1) + \frac{h^2}{12}\left(f^{(3)}(\xi_2) + f^{(3)}(\xi_0)\right).$$

So the **error** is

$$f'(x_1) - D_0 f(x_1) = -\frac{h^2}{12}\big(f^{(3)}(\xi_2) + f^{(3)}(\xi_0)\big).$$

By the intermediate value property for continuous $f^{(3)}$, there exists some $\xi \in (x_0, x_2)$ such that

$$\frac{f^{(3)}(\xi_2) + f^{(3)}(\xi_0)}{2} = f^{(3)}(\xi),$$

so the error can be written as

$$f'(x_1) - D_0 f(x_1) = -\frac{h^2}{6} f^{(3)}(\xi), \quad \xi \in (x_0, x_2).$$

If $|f^{(3)}(x)| \leq M$ for all $x \in [x_0, x_2]$, then

$$|f'(x_1) - D_0 f(x_1)| = \left|\frac{h^2}{6} f^{(3)}(\xi)\right| \leq \frac{M}{6} h^2.$$

So a valid bound is

$$\boxed{|f'(x_1) - D_0 f(x_1)| \leq \frac{M}{6} h^2, \quad M = \max_{x \in [x_0, x_2]} |f^{(3)}(x)|.}$$

Since the leading error term is proportional to h^2, the method is **second-order accurate**:

$$f'(x_1) - D_0 f(x_1) = \mathcal{O}(h^2).$$

So:

- **Error formula (with some $\xi \in (x_0, x_2)$)**:

$$f'(x_1) = D_0 f(x_1) - \frac{h^2}{6} f^{(3)}(\xi).$$

- **Order of accuracy**: 2 (second order).